

Supplementary Material

Physics Is Already There: Rethinking Physical Injection in Latent World Models

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A Training and Evaluation Details

Reproducibility. All training configurations, per-run logs, evaluation protocols, and the exact numbers behind every table and figure are archived alongside the code; the study builds on public assets (LeWM, PhyWorld, Physion, Physion++). We will release the evaluation suite, including the per-horizon/per-partition audit tools and the random-encoder controls.

Hyperparameters. All PhyWorld runs: AdamW, lr 5×10^{-5} (from-scratch reruns: 2×10^{-5}), weight decay 10^{-3} , bf16, gradient clip 1.0, 20 epochs; the from-scratch cells of the pretraining-regime 2×2 train longer and at a lower learning rate so their baselines converge cleanly—60 epochs on uniform motion and collision, 120 on parabola, which needed the extra budget—with injection on and off sharing the identical budget within each domain, so every Δ in Table 2 of the main paper is a within-domain comparison. Batch size 64 for free rollout ($H=8$), 128 for teacher forcing ($H=1$), 48 for $H \geq 16$. History size 3; embedding dimension 192; SIGReg weight 0.09. Seeds: 3072/1234/42 where seed-replicated; both arms of the pretraining-regime 2×2 are three-seed (Table 2 of the main paper): the from-scratch cells were replicated directly, and the fine-tune cells are read off the three-seed scan of Table 1, which is the same regime. Physion++ runs use the readout split (800 clips) with the same optimizer.

OOD partitions. ID: radius/mass $\in [0.7, 1.5]$, velocity $\in [1.0, 4.0]$. OOD draws radius/mass from $[0.3, 0.6] \cup [1.5, 2.0]$ and velocity from $[0, 0.8] \cup [4.5, 6.0]$. Collision uses the four-column variant (two objects’ masses and velocities). Training sets are ID-only (1,000 trajectories); evaluation covers all four partitions ($\geq 1,600$ trajectories per domain).

Probe protocol. Ridge regression ($\alpha=1$), trajectory-level 80/20 splits, probing the encoder output (not the projector), pretrained “paper” initialization, and $K=4$ stacked consecutive latents for velocity. Each choice matters: the naive combination (random init, single frame, through-projector) reads $\rho=0.166$ for uniform velocity where the corrected protocol reads 0.939 (Trap 4, Appendix F). Random-encoder and pixel-statistics controls are run alongside every probe.

From-scratch 2×2 , per seed. Hardest clean split, latent nMSE (uniform/collision both-OOD, parabola r/m-OOD),

seeds 3072/1234/42: uniform off 0.192/0.230/0.246, on 0.750/0.560/0.662; parabola off 0.343/0.420/0.666, on 0.375/0.494/0.473; collision off 0.538/0.513/0.454, on 0.635/0.661/0.649. The parabola from-scratch baseline spans 0.343–0.666 across seeds—a spread wider than the effect being measured, which is why that cell is reported as parity rather than as a gain.

B Full Injection Results

Table 1 is the numeric form of Figure 2 of the main paper, with the seed annotations spelled out.

	uniform (both-OOD)	parabola (r/m-OOD)	collision (both-OOD)
Free-rollout baseline	0.136	0.122	0.479
Fixed slot	0.162°	0.142	0.625
+reweight ($w=30$)	0.132°	0.121°	0.614
+velocity	0.185	0.096	0.630
Probe	0.185	0.137°	0.657
+slot+reweight	0.141°	0.099°	0.623
Kinematic (free MLP)	0.155	0.149°	0.557°
Kinematic $a=g$ (= grounded)	0.173	0.151	0.536°
Consistency	0.143°	0.151	0.610
Label-free prior	0.169	0.120°	0.643

Table 1: The 27-cell scan (hardest clean split, latent nMSE $\downarrow$; every cell and the baseline are three-seed means over seeds 3072/1234/42). **Bold** marks the one seed-robust gain (velocity+slot on parabola: 0.096 ± 0.007 vs. baseline 0.122 ± 0.007 , intervals disjoint). ° marks cells within noise—their seeds overlap the baseline’s; unmarked cells are worse in every seed. 15 cells worse, 11 within noise, 1 better. The $a=g$ row doubles as the grounded prior: adding labels to the label-free structure yields exactly the strict $a=g$ specification, and the configuration was in fact launched under both names before the identity was noticed. The row therefore pools four draws per domain (two independent seed-3072 runs plus the 1234/42 top-ups); the two 3072 draws bracket the mean (uniform 0.206 and 0.166), so the extra weight on that seed does not bias the cell.

Probe/slot factorial (uniform, free rollout). Latent both-OOD nMSE / pixel both-OOD PSNR (dB), seed-3072 val-

ues: baseline 0.131 / 20.41; probe-only 0.167 / 19.83; slot+reweight 0.114 / 21.30; probe+slot+reweight 0.125 / 20.02. The combination underperforms the slot alone on both scales; the latent and pixel verdicts agree.

Slot reweighting sweep (uniform). Slot weight 1/30/100: 0.183 / 0.114 / 0.136 (single seed), with the $w=30$ cell seed-replicated at 0.132 ± 0.017 against the baseline 0.136 ± 0.008 . Kinematic head on the reweighted slot: velocity-OOD decoded position $\rho=0.991/0.987$ (x/y); uniform pixel horizon-28 23.34 vs. 22.09 dB (+1.25); appearance-OOD remains the weak axis (decoded ρ $0.29 \rightarrow 0.43$, vs. 0.96 without the kinematic head).

Redundancy probe details. Protocol for §4.3 of the main paper, which reports the numbers. Two models are probed: the free-rollout baseline (no injection) and the slot-pinned model (structpos, $w=30$). For each, a ridge probe is fit on the dimension subset alone—the full 192, the black box [2:192], the slot [0:2], or a random pair as control—with train/test trajectories split 80/20, so decodability is measured out of sample. Values are Pearson ρ against ground-truth position on the both-OOD partition, averaged over both coordinates, from frame embeddings; the same table holds on rolled-out latents (Appendix D). The slot absorbs position (its supervision loss falls $2.53 \rightarrow 0.015$) without displacing the black box’s redundant copy.

Encoded quantity vs. domain (position monotone, velocity sign-flipping). The reweighted slot’s effect is systematic in what it encodes (hardest clean split, legacy seed-3072 values; the three-seed baselines are 0.136/0.122/0.479, Table 1):

slot content	uniform	parabola	collision
position (structpos)	0.132	0.160	0.596
position+velocity	0.207	0.096	0.621
Δ (adding velocity)	+0.075	−0.064	+0.025

Position is an output variable with the same role in every domain, so pinning it degrades monotonically with dynamics complexity and never flips sign. The marginal effect of adding velocity (+0.075/−0.064/+0.025) tracks velocity’s role in each domain—redundant constant (uniform), linear driver (parabola), discontinuous jump (collision). The parabola gain ($0.122 \pm 0.007 \rightarrow 0.096 \pm 0.007$, every seed lower) is thus the sole seed-robust structural gain in the grid: a fixed form landing on a matching domain, not a portable prior.

Physion++ per-scenario (all ten scenarios, FR at 20 epochs). Rollout nMSE: bouncy_platform 0.041, bouncy_wall 0.094, friction_cloth 0.071, friction_collision 0.062, friction_platform 0.041, mass_collision 0.083, mass_dominoes 0.058, mass_waterpush 0.122; deformable scenes are the common hard case for every method (deform_clothhang 1.375, deform_clothhit 0.772). Injection variants degrade the rigid-body scenarios by 3–10 $\times$ (Table 2) while leaving the deformable failure in place.

Physion++ horizons. Overall nMSE at horizons 16/32/64: baseline ($H=8$) 0.016/0.140/0.280; $H=20$

Scenario	FR	+slot	+cons.	+cons.+acc.
friction_platform	0.041	0.119	0.078	0.067
mass_collision	0.083	0.510	0.682	0.696
mass_dominoes	0.058	0.452	0.586	0.597
mass_waterpush	0.122	0.363	0.663	1.476

Table 2: Physion++ direct training: per-scenario rollout nMSE ($\downarrow$) at 20 epochs. Injection degrades realistic-simulation prediction by 3–10 $\times$ on the mass/friction scenarios shown (deformable-object scenes are the hardest category for all methods, including the baseline). Single-seed; the gaps far exceed baseline seed noise.

0.022/0.033/0.220; $H=16$ +scale 0.010/0.035/0.136; $H=20$ +scale 0.012/0.024/0.087; $H=28$ +scale reaches horizon-64 nMSE 0.014. Free rollout is the main long-horizon lever (horizon-32 cosine 0.91 vs. 0.79–0.87 for every physics-augmented variant); geometric scale jitter acts as an amplitude stabilizer for long rollouts: without it, $H=20$ explodes on friction-collision (nMSE 24.6 at cosine 0.99); with it, 0.019.

C Cross-Backbone Replication: Frozen DINOv2

The cross-backbone variant (§3.5 of the main paper) freezes a DINOv2-small encoder (Oquab et al. 2024)—in the style of DINO-WM (Zhou et al. 2025)—and trains only the projector (as adapter) and the identical predictor stack, losses, and seeds (~ 130 runs). Free-rollout baselines (three-seed): uniform both-OOD 0.427 ± 0.039 , parabola r/m-OOD 0.225 ± 0.019 , collision both-OOD 0.479 ± 0.009 ; Physion++ horizon-64 0.258 ± 0.005 vs. teacher forcing 1.030 ± 0.028 .

Teacher forcing vs. free rollout, per domain. Values behind Figure 4 of the main paper (hardest clean split, latent nMSE $\downarrow$, three-seed mean $\pm$ sample std). LeWM: uniform $0.300 \pm 0.007 \rightarrow 0.136 \pm 0.008$ (2.2 $\times$), parabola $0.443 \pm 0.048 \rightarrow 0.122 \pm 0.007$ (3.6 $\times$), collision $1.152 \pm 0.048 \rightarrow 0.479 \pm 0.079$ (2.4 $\times$), Physion++ horizon-64 $1.174 \pm 0.041 \rightarrow 0.141 \pm 0.018$ (8.3 $\times$). Every teacher-forcing interval is disjoint from its free-rollout counterpart. The frozen-DINOv2 counterparts above give 1.4–4.0 $\times$ over the same four settings, so the lever survives replacing a trainable ViT-tiny with a frozen general-purpose encoder.

Presence and redundancy. The frozen encoder, which has never seen PhyWorld and received no physics supervision or gradient, decodes position from real-frame embeddings at $\rho=0.951$ (both-OOD)—*higher* than the PhyWorld-trained LeWM encoder (0.899). The 190 non-slot dimensions alone decode at 0.951, indistinguishable from the full latent; pinning the slot at weight 300 leaves the black-box copy at 0.973. Presence and the redundant copy are properties of generic visual representations.

Inert, not harmful. Unlike the trainable-encoder regime, almost everything lands within the baseline’s seed band

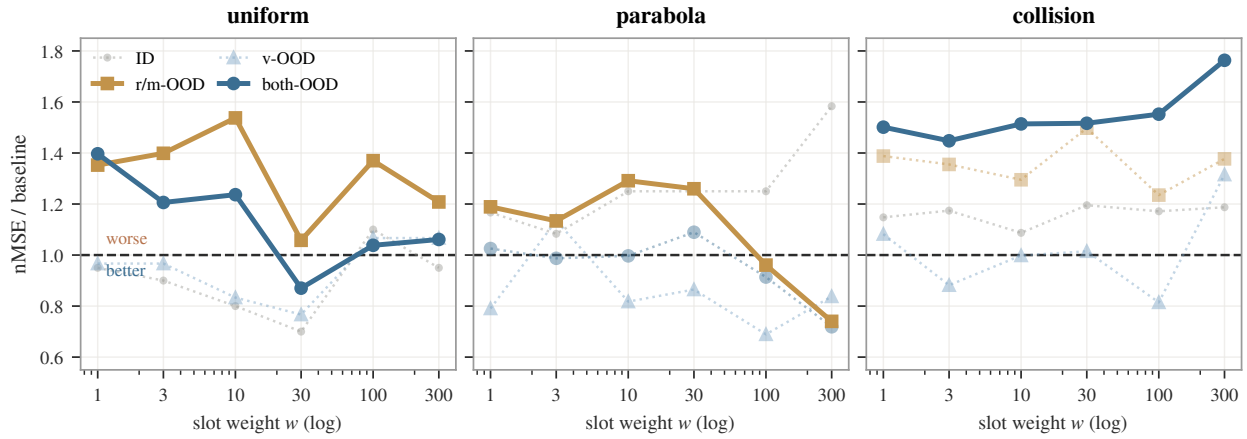


Figure 1: Dose-response re-weighting, all partitions. Solid curves are the four decision cells (hardest clean split); dotted are partitions we do not judge on—ID is not an extrapolation test, and parabola’s v-OOB/both-OOB nMSE is denominator-degenerate (Appendix F). Only 2 of the 4 decision cells reach parity (uniform both-OOB, parabola r/m-OOB); uniform r/m-OOB and collision both-OOB stay above baseline at every weight, and collision degrades as weight grows. The sweep is single-seed; both parity cells were re-run over three seeds and land at baseline (§4.3 of the main paper).

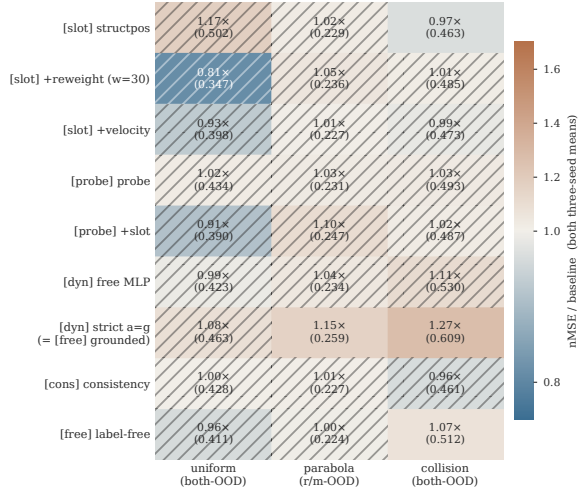


Figure 2: The injection scan repeated on the frozen-DINOv2 backbone, drawn and judged exactly as Figure 2 (three-seed means over three-seed baseline; hatched = the cell’s seeds overlap the baseline’s; the $a=g$ /grounded row merged, four draws). 3 cells worse in every seed / 23 within noise / 1 below (26 of 27 at or below baseline; the one below-cell separates by a 0.001 margin at $0.97\times$). The dynamics-head row is the worst here too ($1.27\times$ on collision); the visually blue uniform cells are all within noise and additionally dissolve under the content-free controls of Trap 5.

(23 of 27 cells): the only cells that separate as worse are the dynamics-head row ($1.15\times$ parabola, $1.27\times$ collision) and the label-free prior on collision ($1.07\times$). With the encoder frozen, injection losses can only reshape the $384 \rightarrow 192$ adapter: the redundant copy cannot be displaced, which bounds the damage—and forecloses the benefit. Injection-variant variance also roughly doubles (± 0.087 vs. baseline

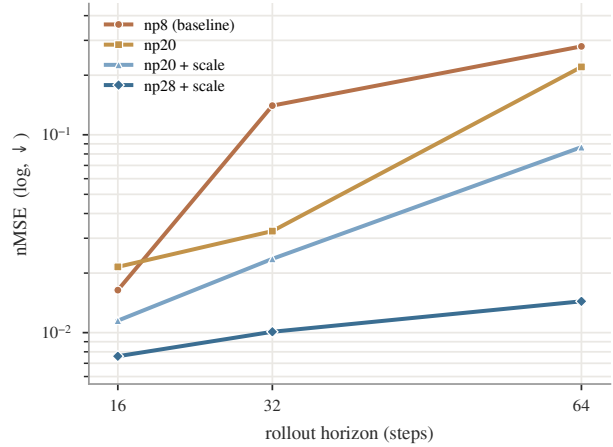


Figure 3: Physion++ direct training: per-horizon rollout nMSE (log scale) as the training horizon grows ($H=8 \rightarrow 28$, +scale jitter). Long-horizon error falls monotonically to $1/19$ of baseline with no inflection: realistic-simulation dynamics reward long-rollout training more than the synthetic domains do.

± 0.039 on uniform), so single-seed cells on this backbone are uninformative.

D Causal Interventions on Trained Models

Protocol and full results for Test 4 (§4.3 of the main paper). All three interventions run at inference on existing checkpoints—no retraining—over three seeds (3072/1234/42), three domains, and both backbones. Position is read with the same ridge probe used elsewhere, fit on the training split.

Why the readout is the black box. Each intervention perturbs one channel and reads position off the *other*. Reading

back the channel just written is partly trivial: the predictor copies its input forward, so the slot readout moves whether or not the perturbation reached the computation. The black-box readout is the one that reflects it (it is also what drives the pixel decoder). Predictions are recorded *before* the per-step perturbation is applied, so every number below is read off what the model itself produced.

Steering. For a target offset δ (in units of position std), we add the minimum-norm latent offset that moves the named channel’s decoded position by δ —for the supervised slot, writing $\text{pos} + \delta$ directly, which is better conditioned—and report $g = \Delta(\text{decoded position})/\delta$. Slot gains on the black-box readout, three-seed mean $\pm$ std, against a norm-matched random direction in the black box:

backbone	domain	g_{slot}	random control
LeWM	uniform	$+1.52 \pm 0.03$	-0.017
LeWM	collision	$+2.01 \pm 0.20$	-0.003
LeWM	parabola	$+1.15 \pm 1.40$	-0.017
DINOv2	uniform	$+1.69 \pm 0.06$	-0.011
DINOv2	parabola	$+0.59 \pm 0.30$	$+0.001$
DINOv2	collision	$+1.94 \pm 0.31$	-0.031

Additivity holds to within 3% ($g_{\text{slot}} + g_{\text{bb}}$ vs. steering both together), so the perturbation stays in the linear regime. *The magnitudes are not proportions*: values above 1 reflect probe $R^2 < 1$, an unperturbed action input, and the 3-frame history anchor. They support “used vs. not” and “this channel vs. that one”, nothing quantitative. LeWM/parabola is excluded: its seeds read $-0.10/+0.90/+2.66$, a spread wider than the effect, and a $20\times$ sweep of δ leaves the per-seed values essentially unchanged, so this is three different models rather than one noisy measurement.

Counterfactual patching. We replace the slot with a donor trajectory’s slot at the same timestep, leaving the black box at its own values, and regress the decoded position on [own truth, donor truth]; the follow-fraction is the donor’s share. The control is the same operation on a model with no injected slot, where those two dimensions carry no physical meaning: LeWM/uniform $0.015 \rightarrow 0.468$, DINOv2/uniform $0.112 \rightarrow 0.650$, DINOv2/collision $0.091 \rightarrow 0.721$. The LeWM parabola and collision controls are not clean (a no-slot model follows $0.42\text{--}0.51$), and we exclude those cells; a collinearity explanation fits parabola (pooled own–donor correlation 0.76) but not collision (0.22), so the cause is unresolved.

Per-dimension Jacobians. $J[c, j] = \partial(\text{decoded position}_c)/\partial(\text{latent}_j)$, averaged over 400 frames, reported per dimension so a 2-d slot and a 190-d black box share a scale. Gradient sensitivity is standard (??), here applied to an intermediate representation rather than to pixels. Unlike the interventions, it chooses no direction, which matters because a redundant code barely moves along any single one; all six cells are usable here where the black-box replacement below leaves one. Its known weakness is the reason we do not rest on it alone: a gradient is local and linear, so it reports what a small change would do rather than what changing the channel

does, and some saliency methods are insensitive to the model they purport to explain (?). The baseline rows answer that second concern directly—the readout tracks whether the model was injected, not the input structure—and the interventions above answer the first. The baseline rows are a built-in control: with no injection those two dimensions are arbitrary and should rank mid-pack, and they do (ratio $0.80\text{--}0.98$, ranks $\sim 90\text{--}162$ of 192). After injection they rank 1st–2nd on LeWM uniform/parabola and DINOv2 uniform, at $2.3\text{--}4.0\times$ the per-dimension sensitivity of the black box; DINOv2/parabola is the exception (ratio 0.99 , ranks $111/90$). The slot nonetheless holds only $0.8\text{--}4.0\%$ of *total* sensitivity—two dimensions against 190—which is the analytic form of “both channels carry load”.

Amnesic projection: a limit we quantify. We remove the linearly position-predictive subspace from the black box by iterative nullspace projection (?), with a rank-matched random-ablation control. Erasing position requires removing 116–154 of the 190 dimensions—the hardest quantification of redundancy in this paper, and far stronger than the random-2-dimension control of Table 3 of the main paper. But at that rank the test loses selectivity: in five of six cells, removing position damages the rollout *less* than removing the same number of dimensions at random (e.g. uniform structpos 0.626 vs. 1.099). The test cannot discriminate in this representation, which follows from how distributed the copy is rather than from how we ran it. Two caveats we state rather than smooth over: the control is matched in rank but not in variance (INLP selects linearly predictive directions, which may be lower-variance than random ones), and an earlier run stopped at 24 removed dimensions showed position-removal damaging the rollout *less* than random removal—apparent support for a bypass—a contrast that disappears once INLP is run to convergence.

Black-box replacement. Replacing k black-box dimensions outright with a donor’s (slot untouched) sidesteps the min-norm problem: on a baseline model, where the black box is the only route position can take, the min-norm patch yields a follow-fraction of 0.089 while replacing all 190 yields 0.961 —an order of magnitude apart on the same model and channel. Sweeping k gives the crossing point at which moving the black box buys as much as moving the 2-d slot. A cell is usable only if its baseline both ignores a patch to the two dimensions it does not have and follows the donor once the whole black box is replaced; only LeWM/uniform passes (0.015 and 0.961), where the crossing sits near 100 dimensions. The remaining five cells fail one or both controls and their crossings are not quoted.

E Augmentation: a Domain-Specific Lever with a Reversal Boundary

Augmentation is not part of the paper’s main line (it is neither an injection mechanism nor a protocol variable), but its boundary behavior is a useful caution. On synthetic domains, per-trial appearance jitter (brightness/contrast, strength 0.5) roughly halves the hardest split: uniform $0.131 \rightarrow 0.068$ (-48%), parabola -63% (r/m-OOD, seed

3072: $0.127 \rightarrow 0.065$); geometric scale jitter, only in combination with a long horizon, sets the collision record (0.172 ± 0.004 over three seeds, vs. the 0.479 ± 0.079 free-rollout baseline). Interactions are non-monotone: appearance conflicts with long-horizon training (0.472, worse than either alone), scale synergizes (0.208), the triple is worse than the best pair (0.253); temporal-stride augmentation loses even to appearance on velocity-OOD (0.556 vs. 0.376), leaving unseen velocities the open hard case.

The appearance recipe reverses on realistic simulation: the same jitter that halves synthetic OOD error degrades Physion++ friction-collision prediction by two orders of magnitude (nMSE $0.062 \rightarrow 6.44$) while the *aggregate* long-horizon cosine improves (Trap 1 below), and it does not help zero-shot transfer either (0.597, below the 0.607 random-prior ceiling). The boundary is principled: appearance jitter encodes an appearance-invariance assumption that holds under PhyWorld’s simple renderer but breaks where appearance carries physical information (friction, mass, material). Scale jitter, which assumes only size-invariance, keeps helping at realistic-simulation scale (Appendix B); appearance jitter flips sign. Augmentation gains are domain- and type-specific, not portable.

F Metric Pathologies and Evaluation Traps

Our anatomy repeatedly reached conclusions opposite to those suggested by common metrics; the decision rules of §4.1 of the main paper distill the following five failure modes, each of which reversed at least one finding.

Trap 1: cosine and probe metrics are duals of the training loss. Probe correlation and latent cosine rise mechanically when probe or slot losses are optimized—they measure whether information is *present*, not whether prediction *uses* it. Probe supervision lifts velocity decodability from $\rho=0.44$ to 0.80 and gives the most stable long-horizon *cosine* (0.882 vs. 0.843), while its *pixel* rollout is the worst in the comparison (19.93 vs. 20.64 dB at horizon 28). Appearance augmentation on Physion++ improves the aggregate long-horizon cosine while per-scene nMSE collapses up to $100\times$ (deform_clothhit: cosine $0.610 \rightarrow 0.913$, nMSE $0.772 \rightarrow 3.69$). Judging either result by direction-only or readout metrics inverts the conclusion. Our own early sweep was misled this way: judged on $K=4$ probe ρ , $\lambda=50$ “won”; re-judged on prediction loss, the ranking reversed.

Trap 2: nMSE denominators can degenerate. When the target sequence degenerates (a parabola ball exits the frame, target variance $\rightarrow 0$), per-horizon nMSE explodes— $3 \times 10^4 - 2 \times 10^6$ at horizon ~ 28 across all six parabola baseline variants, while cosine sits at a healthy 0.55–0.95—and pollutes every aggregate that includes late horizons. This is why parabola is judged on its clean *t/m*-OOD partition throughout. Traps 1 and 2 point in opposite directions (cosine under-reports failure; degenerate nMSE over-reports it); neither metric family is safe alone, hence per-partition, per-horizon cross-validation.

Trap 3: zero-shot transfer is bounded by the architecture prior. On Physion OCP, a *randomly initialized* encoder

scores mean AUC 0.607. No synthetic-training configuration exceeds it: free rollout approaches it (0.603), physics variants fall below it (0.551–0.582), appearance augmentation 0.579–0.597; training longer approaches the ceiling monotonically (0.554 at epoch 1 to 0.603 at epoch 20) without crossing. Per-scenario, only Support (+0.10) and Collide (+0.07) show signal above the random control. Zero-shot claims must be calibrated against the random-prior floor, or they measure the architecture, not the learning.

Trap 4: protocol confounds fabricate results in both directions. Three probe-protocol choices (random vs. pre-trained initialization, single-frame vs. stacked probes, probing through the projector) multiply into a false negative: the naive stack reads uniform velocity decodability at $R^2=0.166$, the corrected protocol 0.939. A silent initialization bug (only 24 of 216 loadable encoder keys actually loaded) invalidated an entire 45-configuration sweep before a loading guard caught it. And converged training loss proves nothing about rollout: our clean from-scratch baseline converges to the pretrained model’s prediction loss (0.006 vs. 0.005) while its rollout OOD error is $2.8\times$ worse.

Trap 5: loss re-weighting masquerades as physics. On the frozen-DINOv2 backbone, heavily re-weighting the pinned position slot (weight 300; the 2 slot dimensions then carry 76% of the prediction loss) improves uniform both-OOD from 0.427 ± 0.039 to 0.286 ± 0.015 —apparently the one clear injection win on either backbone. Two content-free controls dissolve it: pinning the slot to a *shuffled*, physically meaningless target at the same weight reaches 0.330 ± 0.013 , and weakening the pin $10\times$ while keeping the weighting reaches 0.329 ± 0.084 —most of the gain, with no usable physics. All three variants degrade in-distribution error ($0.099 \rightarrow 0.115$ – 0.136), the signature of capacity restriction (the predictor is effectively reduced to a 2-dimensional task), and the same recipe *worsens* collision ($1.11\times$) and parabola ($1.14\times$). An apparent injection gain must be checked against a content-free control at matched loss weighting.

Auxiliary-loss dominance (measurement behind §4.3 of the main paper, Test 2). At probe weight $\lambda=10$, the weighted auxiliary-to-prediction loss ratios—used as a proxy for gradient magnitude; we do not measure gradient norms directly—are $15\times$ (collision), $44\times$ (uniform), and $125\times$ (parabola); the encoder’s effective (participation-ratio) dimensionality collapses by 39–90% relative to baseline (uniform $41 \rightarrow 4$). At $\lambda=1$ the ratios are benign, which is the setting used in §4.2 of the main paper.

Checklist. (i) Report per-partition *and* per-horizon; never aggregate across horizons without inspecting them. (ii) Judge with scale-aware prediction metrics (nMSE, decoded-pixel PSNR); use cosine and probe ρ as diagnostics only. (iii) Audit nMSE denominators for degenerate targets. (iv) Calibrate transfer against a random-init control, and validate probe protocols on a random encoder.

References

Adebayo, J.; Gilmer, J.; Muelly, M.; Goodfellow, I.; Hardt, M.; and Kim, B. 2018. Sanity Checks for Saliency Maps.

In *Advances in Neural Information Processing Systems (NeurIPS)*.

Oquab, M.; Darcet, T.; Moutakanni, T.; Vo, H. V.; Szafraniec, M.; Khalidov, V.; Fernandez, P.; Haziza, D.; Massa, F.; El-Nouby, A.; Assran, M.; Ballas, N.; Galuba, W.; Howes, R.; Huang, P.-Y.; Li, S.-W.; Misra, I.; Rabbat, M.; Sharma, V.; Synnaeve, G.; Xu, H.; Jégou, H.; Mairal, J.; Labatut, P.; Joulin, A.; and Bojanowski, P. 2024. DINOv2: Learning Robust Visual Features without Supervision. *Transactions on Machine Learning Research*.

Ravfogel, S.; Elazar, Y.; Gonen, H.; Twiton, M.; and Goldberg, Y. 2020. Null It Out: Guarding Protected Attributes by Iterative Nullspace Projection. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*.

Simonyan, K.; Vedaldi, A.; and Zisserman, A. 2013. Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps. *arXiv preprint arXiv:1312.6034*.

Sundararajan, M.; Taly, A.; and Yan, Q. 2017. Axiomatic Attribution for Deep Networks. In *International Conference on Machine Learning (ICML)*.

Zhou, G.; Pan, H.; LeCun, Y.; and Pinto, L. 2025. DINO-WM: World Models on Pre-trained Visual Features enable Zero-shot Planning. In *International Conference on Machine Learning (ICML)*.